

Matrix Algebra and Determinants

Introduction: J.J. Sylvester was the first man who introduced the word matrix in 1850 and later on in 1958 Arthur Cayley developed the theory of matrices in a symmetric way. Matrix has wide applications in Science and Engineering.

Definition of Matrix: A matrix is a rectangular array of numbers (real or complex) enclosed by a pair of brackets and the numbers are called the **entries** or the **elements** of the matrix. The general form of a matrix A is written in the following form:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The matrix of m rows and n columns is said to be of order $m \times n$.

We frequently denote such a matrix by simply writing $A = [a_{ij}]$.

Column Matrix: A matrix with only one column is called a column matrix.

Example: 3×1 matrix is a column matrix.

$$\begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}_{(3 \times 1)}$$

Row Matrix: A matrix with only one row is called a row matrix.

Example: 1×3 matrix is a row matrix.

$$[3 \quad 2 \quad 5]_{(1 \times 3)}$$

Equal Matrices: Two matrices are said to be equal if and only if

- (a) they are of the same order, i.e. they have the same number of rows and columns and
- (b) each element of one is equal to the corresponding element of the other.

Given two matrices A and B .

$$A = \begin{bmatrix} a & b \\ d & e \end{bmatrix}, B = \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix}$$

The matrix A is of order 2×2 and B is also of the order 2×2 , so A and B are equal if

$$a = 1, b = -1$$

$$d = 3, e = 2$$

Addition and Scalar Multiplication of Matrices:

Addition of matrices is defined only for the matrices having same number of rows and the same number of columns.

For example if $A = \begin{bmatrix} 1 & -2 \\ 3 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 7 \\ -5 & 4 \end{bmatrix}$ then $A + B = \begin{bmatrix} 3 & 5 \\ -2 & 9 \end{bmatrix}$

The product of the matrix A by a scalar k , written $k.A$ is the matrix obtained by multiplying each element of A by k .

For example if $A = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 4 & 5 \end{bmatrix}$ then $2A = \begin{bmatrix} 2 & -4 & 6 \\ 0 & 8 & 10 \end{bmatrix}$

Matrix Multiplication:

Two matrices A and B are conformable for multiplication if the number of columns in A is equal to the number of rows in B .

Symbolically, let $A = [a_{ij}]$ be the $m \times p$ matrix and $B = [b_{ij}]$ be the $p \times n$ matrix. Then AB is the $m \times n$ matrix $C = [c_{ij}]$ where

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj} = \sum_{k=1}^p a_{ik}b_{kj}$$
$$\begin{cases} i = 1, 2, \dots, m \\ j = 1, 2, \dots, n \end{cases}$$

Example: If $A = \begin{bmatrix} 1 & 3 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 & -4 \\ 5 & -2 & 6 \end{bmatrix}$

Then $AB = \begin{bmatrix} 17 & -6 & 14 \\ -1 & 2 & -14 \end{bmatrix}$

Transpose of a Matrix:

The transpose of a matrix A , written A^T , is the matrix obtained by writing its rows as columns and its columns as rows.

Example: $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

Square Matrix: A matrix with the same number of rows and columns is called a square matrix.

Example: A matrix of the order 3×3 is a square matrix.

$$\begin{bmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \\ 5 & 3 & 4 \end{bmatrix}$$

Diagonal Matrix: A square matrix whose elements $a_{ij} = 0$ when $i \neq j$ is called a diagonal matrix.

Example: $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is a diagonal matrix.

Identity Matrix: A square matrix whose elements $a_{ij} = 0$ if $i \neq j$ and $a_{ij} = 1$ if $i = j$ is called the identity or unit matrix.

Example: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is a 3×3 identity matrix.

Zero Matrix or null matrix: A matrix in which every element is zero is called a null matrix or a zero matrix.

Example: $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is a 3×4 zero matrix.

Sub Matrix: A matrix which is obtained from a given matrix by deleting any number of rows and any number of columns is called a sub matrix of the given matrix.

For example, if we delete first row and first column from the matrix $\begin{bmatrix} 5 & 3 & 2 \\ 1 & 1 & 2 \\ 7 & 3 & 4 \end{bmatrix}$

of order 3×3 , then the matrix so obtained $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is the sub-matrix of the given matrix.

- **Triangular Matrices:** A triangular matrix is a square matrix where all the elements above or below the main diagonal are zero. [↗](#)
 - **Upper Triangular Matrix:** A matrix where all elements below the main diagonal are zero. [↗](#)
 - **Lower Triangular Matrix:** A matrix where all elements above the main diagonal are zero. [↗](#)
- **Strictly Triangular Matrices:** If all the elements on the main diagonal of a (lower or upper) triangular matrix are also zero, the matrix is called strictly (lower or upper) triangular. [↗](#)
 - **Strictly Upper Triangular Matrix:** An upper triangular matrix where all elements on the main diagonal are zero. [↗](#)
 - **Strictly Lower Triangular Matrix:** A lower triangular matrix where all elements on the main diagonal are zero. [↗](#)

- **Example**

Upper Triangular Matrix:

```
1 2 3
0 4 5
0 0 6
```

Strictly Upper Triangular Matrix:

```
0 2 3
0 0 5
0 0 0
```

Lower Triangular Matrix:

```
1 0 0
2 4 0
3 5 6
```

Strictly Lower Triangular Matrix:

```
0 0 0
2 0 0
3 5 0
```

Another Approach:

Upper and lower triangular matrices: Upper triangular matrix is a square matrix in which all the elements below the principal diagonal are zero, i.e., $a_{ij} = 0$ for all $i > j$.

Lower triangular matrix is a square matrix in which all the elements above the principal diagonal are zero, i.e., $a_{ij} = 0$ for all $i < j$.

For **strictly upper triangular matrix** $a_{ij} = 0, \forall i \geq j$ but for a **strictly lower triangular matrix** $a_{ij} = 0, \forall i \leq j$.

Non-singular & Singular Matrix:

A square matrix $D = (d_{ij})$ of order m is said to be non-singular if and only if $|D| \neq 0$ otherwise it is said to be singular.

Example03: $D = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}, |D| = \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} = 0$ i.e. D is singular.

$D = \begin{pmatrix} 1 & -2 \\ 1 & 0 \end{pmatrix}, |D| = \begin{vmatrix} 1 & -2 \\ 1 & 0 \end{vmatrix} = 2$ i.e. D is non-singular.

Minors and Cofactors:

If A is a square matrix, then the *minor of entry a_{ij}* is denoted by M_{ij} and is defined to be the determinant of the sub-matrix that remains after the i th row and j th column are deleted from A . The number $(-1)^{i+j}M_{ij}$ is denoted by C_{ij} and is called the *cofactor of entry a_{ij}* .

Example04: Let $A = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$. The minor of entry a_{11} is $M_{11} = \begin{vmatrix} 2 & 5 & 6 \\ 1 & 4 & 8 \end{vmatrix} = \begin{vmatrix} 5 & 6 \\ 4 & 8 \end{vmatrix} = 16$.

The cofactor of a_{11} is $C_{11} = (-1)^{1+1}M_{11} = M_{11} = 16$.

Similarly, the minor of entry a_{32} is $M_{32} = \begin{vmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{vmatrix} = \begin{vmatrix} 3 & -4 \\ 2 & 6 \end{vmatrix} = 26$.

The cofactor of a_{32} is $C_{32} = (-1)^{3+2}M_{32} = -M_{32} = -26$.

Adjoint of a Matrix:

If A is any $n \times n$ matrix and C_{ij} is the cofactor of a_{ij} , then the matrix

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}$$

is called the *matrix of cofactors from A* . The transpose of this matrix is called the *adjoint of A* and is denoted by $\text{adj}(A)$.

Theorem: Inverse of a Matrix Using Its Adjoint

If A is an invertible matrix, then

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

Inverse of a Matrix: If A is a square matrix, and if a matrix B of the same size can be found such that $AB = BA = I$, then A is said to be invertible and B is called an inverse of A . If no such matrix B can be found, then A is said to be singular.

The matrix $B = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix}$ is an inverse of $A = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix}$

$$\text{since } AB = \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$\text{and } BA = \begin{bmatrix} 3 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -5 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

A Matrix with no Inverse

The matrix $A = \begin{bmatrix} 1 & 4 & 0 \\ 2 & 5 & 0 \\ 3 & 6 & 0 \end{bmatrix}$ is singular. To see why, let

$$B = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$$

be any 3×3 matrix. The third column of BA is

$$\begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ thus } BA \neq I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Theorem

The matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible if $ad - bc \neq 0$, in which case the inverse is given by the formula

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example: Consider the matrix $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ then

$$A^{-1} = \frac{1}{1} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$

Row operations of Matrix:

The matrix operations of

1. Interchanging two rows,
2. Adding a multiple of one row to another,
3. Multiplying any row by a nonzero element.

Elementary Matrix: An $n \times n$ matrix is called an elementary matrix if it can be obtained from the $n \times n$ identity matrix I_n by performing a single row operation.

Example: $\begin{bmatrix} 1 & 0 \\ 0 & -3 \end{bmatrix}$ (Multiply the second row of I_2 by -3).

$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (Add 3 times the third row of I_3 to the first row)

Problems and Exercise

Type-1: Equal Matrices

Problem-1:

If

$$\begin{bmatrix} a+b & 2 \\ 5 & ab \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$$

find the values of a and b .

Solution:

Since two matrices are equal, their corresponding elements must be equal. Comparing the elements from both matrices, we obtain the following equations:

1. From the first row, first column:

$$a + b = 6$$

2. From the second row, second column:

$$ab = 8$$

Now, we solve for a and b .

Step 1: Express b in Terms of a

From the first equation:

$$b = 6 - a$$

Step 2: Substitute into the Second Equation

$$a(6 - a) = 8$$

$$6a - a^2 = 8$$

$$a^2 - 6a + 8 = 0$$

Step 3: Solve the Quadratic Equation

$$(a - 2)(a - 4) = 0$$

Thus, the possible values for a are:

$$a = 2 \quad \text{or} \quad a = 4$$

Step 4: Find Corresponding Values of b

For $a = 2$:

$$b = 6 - 2 = 4$$

For $a = 4$:

$$b = 6 - 4 = 2$$

Final Answer:

The possible values of (a, b) are:

$$(2, 4) \quad \text{or} \quad (4, 2)$$

Problem-2:

If

$$\begin{bmatrix} x+y & 4 \\ 3 & xy \end{bmatrix} = \begin{bmatrix} 9 & 4 \\ 3 & 14 \end{bmatrix}$$

find the values of x and y .

Solution:

Equating corresponding elements:

1. $x + y = 9$

2. $xy = 14$

Substituting $y = 9 - x$ into $xy = 14$:

$$x(9 - x) = 14$$

$$x^2 - 9x + 14 = 0$$

Factoring:

$$(x - 7)(x - 2) = 0$$

So, $x = 7$ or $x = 2$, giving $y = 2$ or $y = 7$.

Final Answer:

$$(7, 2) \quad \text{or} \quad (2, 7)$$

Exercise Set:

Problem-1:

If

$$\begin{bmatrix} a+b & 5 \\ 4 & ab \end{bmatrix} = \begin{bmatrix} 10 & 5 \\ 4 & 21 \end{bmatrix}$$

find the values of a and b .

Problem-2:

If

$$\begin{bmatrix} x-y & 6 \\ 2 & xy \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 2 & 8 \end{bmatrix}$$

find the values of x and y .

Problem-3:

If

$$\begin{bmatrix} p+q & 7 \\ 5 & pq \end{bmatrix} = \begin{bmatrix} 12 & 7 \\ 5 & 20 \end{bmatrix}$$

find the values of p and q .

Problem-4:

If

$$\begin{bmatrix} m+n & 8 \\ 3 & mn \end{bmatrix} = \begin{bmatrix} 11 & 8 \\ 3 & 18 \end{bmatrix}$$

find the values of m and n .

Type-2: Matrix Equations

Problem-1:

Find x, y, z, t if

$$2 \begin{bmatrix} x & z \\ y & t \end{bmatrix} + 3 \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} = 3 \begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$$

Solution:

The given matrix equation can be rewritten as:

$$\begin{bmatrix} 2x & 2z \\ 2y & 2t \end{bmatrix} + \begin{bmatrix} 3 & -3 \\ 0 & 6 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 12 & 18 \end{bmatrix}$$

Subtracting the second matrix from the third:

$$\begin{bmatrix} 2x + 3 & 2z - 3 \\ 2y & 2t + 6 \end{bmatrix} = \begin{bmatrix} 9 & 15 \\ 12 & 18 \end{bmatrix}$$

Equating corresponding elements:

1. $2x + 3 = 9$
2. $2z - 3 = 15$
3. $2y = 12$
4. $2t + 6 = 18$

Solving for x, y, z, t :

$$x = \frac{9 - 3}{2} = 3, \quad z = \frac{15 + 3}{2} = 9, \quad y = \frac{12}{2} = 6, \quad t = \frac{18 - 6}{2} = 6$$

Final Answer:

$$x = 3, \quad z = 9, \quad y = 6, \quad t = 6$$

Problem-2:

Find X and Y if

$$X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}, \quad X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}.$$

Solution:

We have the given equations:

$$X + Y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}, \quad X - Y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}.$$

Adding both equations:

$$\begin{aligned} (X + Y) + (X - Y) &= \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ 2X &= \begin{bmatrix} 7+3 & 0+0 \\ 2+0 & 5+3 \end{bmatrix} = \begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix} \end{aligned}$$

Dividing by 2:

$$X = \frac{1}{2} \begin{bmatrix} 10 & 0 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}$$

Subtracting the second equation from the first:

$$\begin{aligned} (X + Y) - (X - Y) &= \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix} - \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \\ 2Y &= \begin{bmatrix} 7-3 & 0-0 \\ 2-0 & 5-3 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 2 & 2 \end{bmatrix} \end{aligned}$$

Dividing by 2:

$$Y = \frac{1}{2} \begin{bmatrix} 4 & 0 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

Final Answer:

$$X = \begin{bmatrix} 5 & 0 \\ 1 & 4 \end{bmatrix}, \quad Y = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$$

Problem-3:

If

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 0 \\ -2 & 7 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & -2 & 7 \\ 8 & 0 & 9 \\ 3 & 1 & -5 \end{bmatrix}$$

find a matrix X such that

$$2A + 3X = 5B.$$

Solution:

Given equation:

$$2A + 3X = 5B$$

Rearrange:

$$3X = 5B - 2A$$

Compute $5B$:

$$5B = 5 \times \begin{bmatrix} 6 & -2 & 7 \\ 8 & 0 & 9 \\ 3 & 1 & -5 \end{bmatrix} = \begin{bmatrix} 30 & -10 & 35 \\ 40 & 0 & 45 \\ 15 & 5 & -25 \end{bmatrix}$$

Compute $2A$:

$$2A = 2 \times \begin{bmatrix} 2 & -1 & 3 \\ 4 & 2 & 0 \\ -2 & 7 & 3 \end{bmatrix} = \begin{bmatrix} 4 & -2 & 6 \\ 8 & 4 & 0 \\ -4 & 14 & 6 \end{bmatrix}$$

Subtract $2A$ from $5B$:

$$3X = \begin{bmatrix} 30 & -10 & 35 \\ 40 & 0 & 45 \\ 15 & 5 & -25 \end{bmatrix} - \begin{bmatrix} 4 & -2 & 6 \\ 8 & 4 & 0 \\ -4 & 14 & 6 \end{bmatrix}$$
$$3X = \begin{bmatrix} 26 & -8 & 29 \\ 32 & -4 & 45 \\ 19 & -9 & -31 \end{bmatrix}$$

Divide by 3:

$$X = \frac{1}{3} \begin{bmatrix} 26 & -8 & 29 \\ 32 & -4 & 45 \\ 19 & -9 & -31 \end{bmatrix}$$
$$X = \begin{bmatrix} 26/3 & -8/3 & 29/3 \\ 32/3 & -4/3 & 45/3 \\ 19/3 & -9/3 & -31/3 \end{bmatrix}$$

Final Answer:

$$X = \begin{bmatrix} 26/3 & -8/3 & 29/3 \\ 32/3 & -4/3 & 15 \\ 19/3 & -3 & -31/3 \end{bmatrix}$$

Exercise Set: Matrix Equations

Problem-1:

If

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$$

find X such that

$$2A + 3X = 4B.$$

Problem-2:

Find X and Y if

$$X + Y = \begin{bmatrix} 9 & 4 \\ 3 & 6 \end{bmatrix}, \quad X - Y = \begin{bmatrix} 1 & 2 \\ 5 & 0 \end{bmatrix}.$$

Problem-3:

If

$$A = \begin{bmatrix} 2 & -3 & 4 \\ 1 & 5 & -2 \\ -4 & 0 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & -2 & 8 \\ 4 & 0 & 10 \\ -2 & 7 & 5 \end{bmatrix}$$

find X such that

$$3A + 2X = 5B.$$

Problem-4:

Solve for X if

$$AX + B = C,$$

where

$$A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ -2 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 5 & 3 \\ 7 & 8 \end{bmatrix}.$$

Problem-5:

Find X such that

$$2X - A = B,$$

where

$$A = \begin{bmatrix} 3 & 5 \\ -1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 7 & 9 \\ 2 & 6 \end{bmatrix}.$$

Type-3: Matrix Multiplication

Problem-1:

Show that

$$A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 2 & 4 \\ 0 & 0 & 2 \end{bmatrix}$$

satisfies the equation

$$A^2 - 3A + 2I_3 = 0$$

Solution:

Step 1: Compute $A^2 = A \cdot A$

$$A^2 = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 2 & 4 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & -2 \\ 2 & 2 & 4 \\ 0 & 0 & 2 \end{bmatrix}$$

Computing the entries:

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 \cdot 1 + 0 \cdot 2 + (-2) \cdot 0 & 1 \cdot 0 + 0 \cdot 2 + (-2) \cdot 0 & 1 \cdot (-2) + 0 \cdot 4 + (-2) \cdot 2 \\ 2 \cdot 1 + 2 \cdot 2 + 4 \cdot 0 & 2 \cdot 0 + 2 \cdot 2 + 4 \cdot 0 & 2 \cdot (-2) + 2 \cdot 4 + 4 \cdot 2 \\ 0 \cdot 1 + 0 \cdot 2 + 2 \cdot 0 & 0 \cdot 0 + 0 \cdot 2 + 2 \cdot 0 & 0 \cdot (-2) + 0 \cdot 4 + 2 \cdot 2 \end{bmatrix} \\ &= \begin{bmatrix} 1 + 0 + 0 & 0 + 0 + 0 & -2 + 0 - 4 \\ 2 + 4 + 0 & 0 + 4 + 0 & -4 + 8 + 8 \\ 0 + 0 + 0 & 0 + 0 + 0 & 0 + 0 + 4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & -6 \\ 6 & 4 & 12 \\ 0 & 0 & 4 \end{bmatrix} \end{aligned}$$

Step 2: Compute $3A$

$$\begin{aligned} 3A &= 3 \times \begin{bmatrix} 1 & 0 & -2 \\ 2 & 2 & 4 \\ 0 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 3 & 0 & -6 \\ 6 & 6 & 12 \\ 0 & 0 & 6 \end{bmatrix} \end{aligned}$$

Step 3: Compute $2I_3$

$$\begin{aligned} 2I_3 &= 2 \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{aligned}$$

Step 4: Compute $A^2 - 3A + 2I_3$

$$\begin{aligned} A^2 - 3A + 2I_3 &= \begin{bmatrix} 1 & 0 & -6 \\ 6 & 4 & 12 \\ 0 & 0 & 4 \end{bmatrix} - \begin{bmatrix} 3 & 0 & -6 \\ 6 & 6 & 12 \\ 0 & 0 & 6 \end{bmatrix} + \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 1-3+2 & 0+0+0 & -6+6+0 \\ 6-6+0 & 4-6+2 & 12-12+0 \\ 0-0+0 & 0-0+0 & 4-6+2 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \end{aligned}$$

Thus,

$$A^2 - 3A + 2I_3 = 0.$$

Final Conclusion:

Hence, the given matrix A satisfies the equation $A \downarrow 3A + 2I_3 = 0$ as required.

Problem-2:

If

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}$$

and

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

then find k such that

$$A^2 = 8A + kI.$$

Solution:

We have

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}.$$

Step 1: Compute $A^2 = A \cdot A$

$$A^2 = \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix}$$

Computing the entries:

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 \cdot 1 + 0 \cdot (-1) & 1 \cdot 0 + 0 \cdot 7 \\ -1 \cdot 1 + 7 \cdot (-1) & -1 \cdot 0 + 7 \cdot 7 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ -1 - 7 & 49 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ -8 & 49 \end{bmatrix}. \end{aligned}$$

Step 2: Compute $8A$

$$\begin{aligned} 8A &= 8 \times \begin{bmatrix} 1 & 0 \\ -1 & 7 \end{bmatrix} \\ &= \begin{bmatrix} 8 & 0 \\ -8 & 56 \end{bmatrix}. \end{aligned}$$

Step 3: Compute kI

$$\begin{aligned} kI &= k \times \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}. \end{aligned}$$

Step 4: Compute $8A + kI$

$$\begin{aligned} 8A + kI &= \begin{bmatrix} 8 & 0 \\ -8 & 56 \end{bmatrix} + \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \\ &= \begin{bmatrix} 8 + k & 0 \\ -8 + 0 & 56 + k \end{bmatrix} \\ &= \begin{bmatrix} 8 + k & 0 \\ -8 & 56 + k \end{bmatrix}. \end{aligned}$$

Step 5: Solve for k

Since

$$A^2 = 8A + kI,$$

we equate the corresponding entries:

1. $1 = 8 + k$

$$k = 1 - 8 = -7$$

2. $56 + k = 49$

$$k = 49 - 56 = -7$$

Thus, solving the equations, we get

$$k = -7.$$

Problem-3:

Consider the matrices:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 \end{bmatrix}.$$

Verify that

$$(AB)C = A(BC).$$

Solution:

Step 1: Compute AB

Since A is 2×3 and B is 3×1 , their product AB is a 2×1 matrix:

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Computing the entries:

$$\begin{aligned} AB &= \begin{bmatrix} (1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3) \\ (4 \cdot 1 + 5 \cdot 2 + 6 \cdot 3) \end{bmatrix} \\ &= \begin{bmatrix} 1 + 4 + 9 \\ 4 + 10 + 18 \end{bmatrix} \\ &= \begin{bmatrix} 14 \\ 32 \end{bmatrix}. \end{aligned}$$

Step 2: Compute $(AB)C$

Since AB is 2×1 and C is 1×2 , their product $(AB)C$ is a 2×2 matrix:

$$(AB)C = \begin{bmatrix} 14 \\ 32 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix}$$

Computing the entries:

$$\begin{aligned} (AB)C &= \begin{bmatrix} 14 \cdot 1 & 14 \cdot 2 \\ 32 \cdot 1 & 32 \cdot 2 \end{bmatrix} \\ &= \begin{bmatrix} 14 & 28 \\ 32 & 64 \end{bmatrix}. \end{aligned}$$

Step 3: Compute BC

Since B is 3×1 and C is 1×2 , their product BC is a 3×2 matrix:

$$BC = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix}$$

Computing the entries:

$$\begin{aligned} BC &= \begin{bmatrix} 1 \cdot 1 & 1 \cdot 2 \\ 2 \cdot 1 & 2 \cdot 2 \\ 3 \cdot 1 & 3 \cdot 2 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}. \end{aligned}$$

Step 4: Compute $A(BC)$

Since A is 2×3 and BC is 3×2 , their product $A(BC)$ is a 2×2 matrix:

$$A(BC) = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{bmatrix}$$

Computing the entries:

$$\begin{aligned} A(BC) &= \begin{bmatrix} (1 \cdot 1 + 2 \cdot 2 + 3 \cdot 3) & (1 \cdot 2 + 2 \cdot 4 + 3 \cdot 6) \\ (4 \cdot 1 + 5 \cdot 2 + 6 \cdot 3) & (4 \cdot 2 + 5 \cdot 4 + 6 \cdot 6) \end{bmatrix} \\ &= \begin{bmatrix} 1 + 4 + 9 & 2 + 8 + 18 \\ 4 + 10 + 18 & 8 + 20 + 36 \end{bmatrix} \\ &= \begin{bmatrix} 14 & 28 \\ 32 & 64 \end{bmatrix}. \end{aligned}$$

Final Conclusion:

Since

$$(AB)C = A(BC) = \begin{bmatrix} 14 & 28 \\ 32 & 64 \end{bmatrix},$$

the given property holds true.

Exercise Set

Problem 1:

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Show that A satisfies the equation

$$A^2 - 3A + I_2 = 0.$$

Problem 2:

If

$$A = \begin{bmatrix} 0 & 2 \\ -1 & 3 \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

find k such that

$$A^2 = 5A + kI.$$

Problem 3:

Consider the matrices:

$$A = \begin{bmatrix} 3 & 1 & 4 \\ 2 & 0 & 5 \end{bmatrix}, B = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}, C = \begin{bmatrix} 2 & -1 \end{bmatrix}.$$

Verify that

$$(AB)C = A(BC).$$

Problem 4:

Let

$$A = \begin{bmatrix} 3 & 0 & -2 \\ 1 & 4 & 2 \\ 0 & -1 & 5 \end{bmatrix}.$$

Show that A satisfies the equation

$$A^2 - 6A + 7I_3 = 0.$$

Problem 5:

If

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & -2 \\ 1 & 5 & 6 \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

find k such that

$$A^2 = 7A + kI.$$

Problem 6:

Consider the matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}, C = \begin{bmatrix} 4 \\ -2 \end{bmatrix}.$$

Verify that

$$(AB)C = A(BC).$$

Type-4: Matrix Inversion using Adjoint

Problem-1: Find the Inverse of the following Matrix using Adjoint.

$$A = \begin{bmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{bmatrix}$$

Solution:

The determinant of A is

$$\begin{aligned} \det(A) &= \begin{vmatrix} 3 & 2 & -1 \\ 1 & 6 & 3 \\ 2 & -4 & 0 \end{vmatrix} \\ &= 3(0 + 12) - 2(0 - 6) - 1(-4 - 12) \\ &= 36 + 12 + 16 = 64 \end{aligned}$$

The cofactors of A are

$$C_{11} = 12, \quad C_{12} = 6, \quad C_{13} = -16$$

$$C_{21} = 4, \quad C_{22} = 2, \quad C_{23} = 16$$

$$C_{31} = 12, \quad C_{32} = -10, \quad C_{33} = 16$$

So the cofactor matrix is

$$C = \begin{bmatrix} 12 & 6 & -16 \\ 4 & 2 & 16 \\ 12 & -10 & 16 \end{bmatrix}$$

Since the adjoint of A is the transpose of the cofactor matrix, we write

$$\text{adj}(A) = C^T = \begin{bmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{bmatrix}$$

Now, the inverse of A is

$$\begin{aligned} A^{-1} &= \frac{1}{\det(A)} C^T = \frac{1}{64} \begin{bmatrix} 12 & 4 & 12 \\ 6 & 2 & -10 \\ -16 & 16 & 16 \end{bmatrix} \\ &= \begin{bmatrix} \frac{12}{64} & \frac{4}{64} & \frac{12}{64} \\ \frac{6}{64} & \frac{2}{64} & \frac{-10}{64} \\ \frac{-16}{64} & \frac{16}{64} & \frac{16}{64} \end{bmatrix} \end{aligned}$$

Problem-2: Find the Inverse of the following Matrix using Adjoint.

$$B = \begin{bmatrix} 4 & 3 & 2 \\ 2 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix}$$

Solution:

The determinant of B is

$$\begin{aligned} \det(B) &= \begin{vmatrix} 4 & 3 & 2 \\ 2 & 2 & 1 \\ 1 & 3 & 2 \end{vmatrix} \\ &= 3 \end{aligned}$$

The cofactors of B are

$$C_{11} = 1, \quad C_{12} = -3, \quad C_{13} = 4$$

$$C_{31} = -1, \quad C_{32} = 2, \quad C_{33} = 2$$

$$C_{21} = -1, \quad C_{22} = 2, \quad C_{23} = -2$$

So the cofactor matrix is

$$C = \begin{bmatrix} 1 & -3 & 4 \\ -1 & 2 & -2 \\ -1 & 2 & 2 \end{bmatrix}$$

Since the adjoint of B is the transpose of the cofactor matrix, we write

$$\text{adj}(B) = C^T = \begin{bmatrix} 1 & -1 & -1 \\ -3 & 2 & 2 \\ 4 & -2 & 2 \end{bmatrix}$$

Now, the inverse of B is

$$B^{-1} = \frac{1}{\det(B)} C^T = \frac{1}{3} \begin{bmatrix} 1 & -1 & -1 \\ -3 & 2 & 2 \\ 4 & -2 & 2 \end{bmatrix}$$

Exercise Set: Matrix Inversion Using the Adjoint Method

Conceptual Questions

1. Explain the steps involved in finding the inverse of a square matrix using the adjoint method.
2. What is the condition for a matrix to be invertible using the adjoint method?
3. Why is the determinant important in the adjoint method of inversion?

Numerical Problems

Problem 1: 2×2 Matrix Inversion

Find the inverse of the matrix using the adjoint method:

$$A = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$$

Problem 2: 3×3 Matrix Inversion

Find the inverse of the following matrix using the adjoint method:

$$B = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 0 & 2 \\ 4 & 1 & 5 \end{bmatrix}$$

Problem 3: Determining Invertibility

Check if the following matrix is invertible using the adjoint method. If it is, find its inverse:

$$C = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Problem 4: Special Case (Singular Matrix)

Consider the matrix

$$D = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

- (a) Compute its determinant.
- (b) Explain why the adjoint method fails for this matrix.

Problem 5: Adjoint Calculation

Find the adjoint of the matrix:

$$E = \begin{bmatrix} 3 & 0 & 2 \\ 2 & 0 & -2 \\ 0 & 1 & 1 \end{bmatrix}$$

Then, use it to compute E^{-1} .

Type-6: Using Row Operations, find the Inverse of a Matrix

Example 01: Find the inverse of

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

Solution: We want to reduce A to the identity matrix by row operations and simultaneously apply these operations to I to produce A^{-1} . To accomplish this we shall adjoin the identity matrix to the right side of A , thereby producing a matrix of the form:

$$[A|I]$$

Then we shall apply row operations so that the final matrix will have the form:

$$[I|A^{-1}]$$

The computations are as follows:

$$\begin{aligned} [A|I] &= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right] \\ &= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right] \\ &\quad [R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - R_1] \\ &= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] \\ &\quad [R_3 \rightarrow 2R_2 + R_3] \\ &= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \\ &\quad [R_3 \rightarrow -R_3] \end{aligned}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right]$$

$$[R_1 \rightarrow R_1 - 3R_3, R_2 \rightarrow 3R_3 + R_2]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right]$$

$$[R_1 \rightarrow R_1 - 2R_2]$$

$$= [I|A^{-1}]$$

Thus,

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

Exercise Problems:

1. Find the inverse of the following matrix by using row canonical form:

$$A = \begin{bmatrix} 3 & 4 & -1 \\ 1 & 0 & 3 \\ 2 & 5 & -4 \end{bmatrix} \quad \text{Answer: } A^{-1} = \begin{bmatrix} \frac{3}{2} & -\frac{11}{10} & -\frac{6}{5} \\ -1 & 1 & 1 \\ -\frac{1}{2} & \frac{7}{10} & \frac{2}{5} \end{bmatrix}$$

2. Find the inverse of the following matrix:

$$A = \begin{bmatrix} 2 & -1 & -1 \\ 1 & -2 & 1 \\ 1 & -1 & 2 \end{bmatrix} \quad \text{Answer: } A^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{6} & -\frac{5}{6} & \frac{1}{2} \\ -\frac{1}{6} & -\frac{1}{6} & \frac{1}{2} \end{bmatrix}$$